

LAB 1

REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

Problem Statement #09

FACULTY RESEARCH GRANT & PUBLICATION TRACKER

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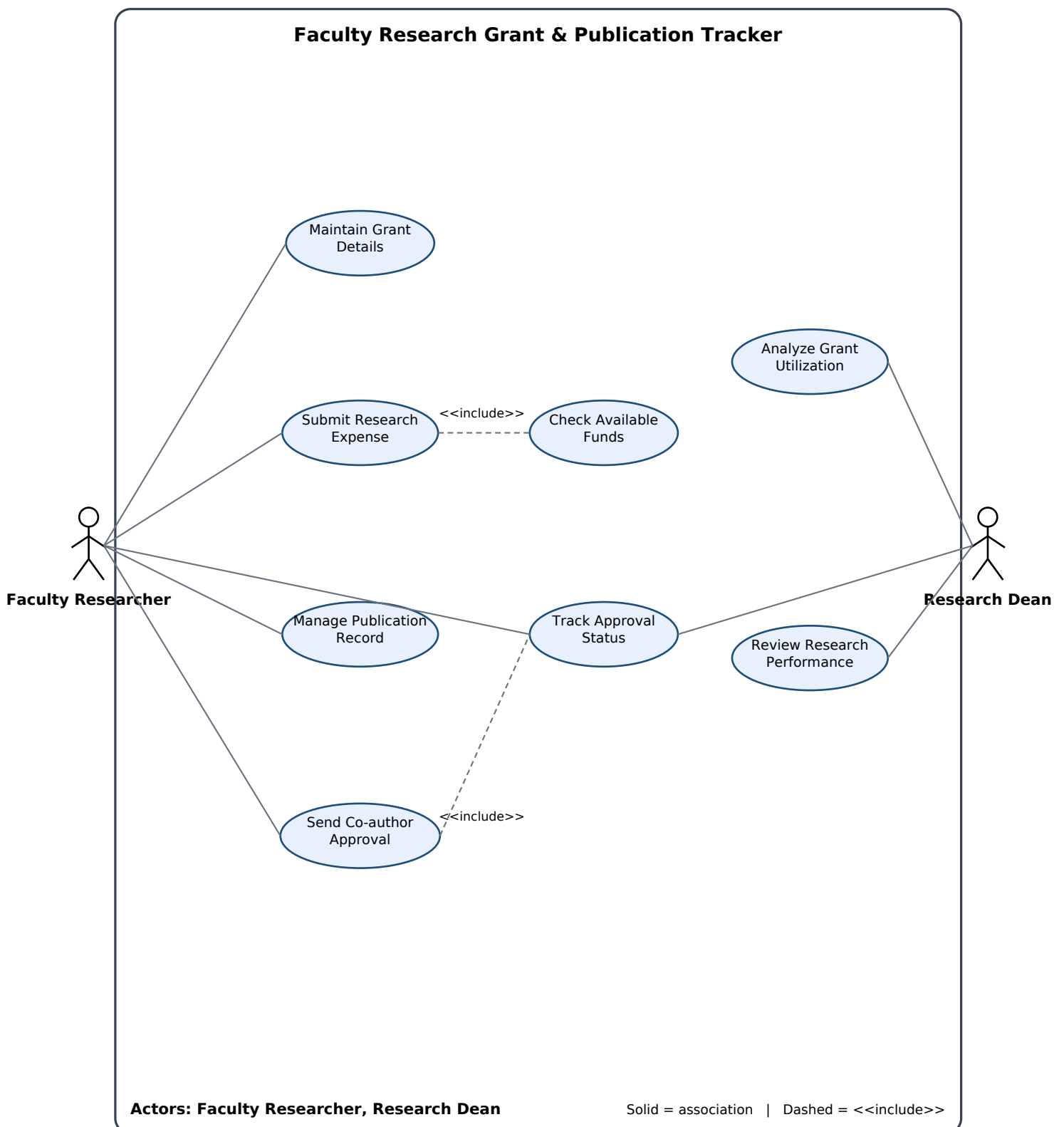
Requirements, UML Use-Case Model, Use-Case Catalogue and Core Use-Case Specification

1. Complete Requirements Table

ID	Type	Requirement	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow Faculty Researchers to create and maintain research grant records, including grant allocation and approved funding limits.	High	Pass: A researcher can create/update a grant record and the system displays the correct allocated amount and funding limit. Fail: Grant details are missing, inconsistent, or cannot be updated.	Provides a central record of sponsored research grants and their approved funding.
FR-002	Functional	The system shall allow Faculty Researchers to record equipment procurement and other research expenses, calculate the remaining grant balance, and block expenses that exceed the grant cap.	High	Pass: A valid expense updates the remaining balance accurately; an expense exceeding the available grant amount is rejected. Fail: The system allows an overdraft or calculates the balance incorrectly.	Prevents grant overspending and keeps financial information accurate.
FR-003	Functional	The system shall allow Faculty Researchers to record publications and maintain publication metrics including indexing status and citation information.	High	Pass: A publication can be recorded and its indexing/citation information can be stored and updated. Fail: Publication records cannot be maintained or metrics are not reflected correctly.	Enables centralized tracking of faculty publication output and impact.
FR-004	Functional	The system shall support a co-author approval workflow for publications and allow the approval status to be tracked.	Medium	Pass: A publication can be submitted for co-author approval and its status changes according to approval actions. Fail: Approval requests cannot be submitted or status is not tracked.	Provides a structured workflow for obtaining and monitoring co-author approvals.
FR-005	Functional	The system shall provide the Research Dean with fund burn-up analytics and research/publication performance information.	Medium	Pass: The Research Dean can view grant fund utilization and relevant publication metrics. Fail: Analytics are unavailable, incomplete, or inconsistent with stored records.	Helps the Research Dean monitor sponsored-fund utilization and faculty research performance.
NFR-001	Non-Functional	The system shall maintain an immutable, tamper-evident audit ledger for financial approvals and journal/publication status modifications.	High	Pass: Audit records cannot be silently altered or deleted and changes to financial approvals and publication status are traceable. Fail: A user can modify or remove audit history without detection.	Protects the integrity and accountability of financial and publication workflow records.
NFR-002	Non-Functional	The system shall provide responsive access to grant, expense, publication, approval, and analytics operations under expected peak usage.	High	Pass: Benchmarking under simulated peak load demonstrates acceptable response performance without service degradation. Fail: Response times become unacceptable or operations fail under expected load.	Ensures the portal remains usable when multiple faculty researchers and administrators access it concurrently.

2. UML Use-Case Diagram

Actors: Faculty Researcher and Research Dean.



3. Use Cases

Use Case	Primary Actor	Purpose	Related Requirement(s)
UC-01 — Maintain Grant Details	Faculty Researcher	Create or update a sponsored grant record and keep its approved ceiling, dates and status current.	FR-01
UC-02 — Submit Research Expense	Faculty Researcher	Enter a research-related purchase or expense against an active grant.	FR-02
UC-03 — Check Available Funds	Included behaviour	Determine the remaining amount after considering approved expenditure for the selected grant.	FR-02
UC-04 — Manage Publication Record	Faculty Researcher	Add and maintain bibliographic information and research-impact fields for a publication.	FR-03
UC-05 — Send Co-author Approval	Faculty Researcher	Route a publication to its co-authors and record the approval request.	FR-04
UC-06 — Track Approval Status	Faculty Researcher / Research Dean	View whether a publication is pending, approved or requires action.	FR-04
UC-07 — Analyze Grant Utilization	Research Dean	Review spending against approved grant limits and identify remaining funds.	FR-05
UC-08 — Review Research Performance	Research Dean	Inspect publication activity, indexing information and other stored research indicators.	FR-05

4. Core Use-Case Specification — Submit Research Expense

Primary Actor: Faculty Researcher

Preconditions: The researcher has signed in, an active grant is available, and the researcher is authorized to enter expenses.

Postconditions: For a valid submission, the expense is stored, expenditure totals are updated and the remaining amount is recalculated. For an invalid submission, no financial record is created.

Trigger: The researcher needs to record a new project expense.

Main Success Scenario

1. The researcher signs in to the portal.
2. The portal displays grants that the researcher can access.
3. The researcher chooses an active grant.
4. The researcher opens the expense-entry option.
5. The portal displays fields for category, description, date and amount.
6. The researcher enters the transaction details.
7. The researcher submits the transaction.
8. The portal checks that required fields and the amount are valid.
9. The portal determines the grant's available funds.
10. The portal verifies that the new expense does not cross the approved ceiling.
11. The transaction is stored successfully.
12. The portal recalculates total spending and the remaining amount.
13. A confirmation message and updated balance are shown to the researcher.